# GitHub Workflow: Cars CRUD Testing Exercise

**Goal:** Publish the Cars CRUD app, write and run tests on a separate branch, and merge them into `main` through a pull request.

## 1. Create a GitHub repository

1. On GitHub, select **New repository** (https://github.com/new).
2. Name it `cars-crud`, choose **public** visibility, and click **Create repository**.
3. Leave **Add a README**, **.gitignore**, and **license** unchecked (these files already exist locally).

## 2. Initialize Git and push the app

In a terminal, open the `cars-crud` directory (the one containing `app/`, `tests/`, and `requirements-dev.txt`):

```
cd cars-crud
git init -b main
git add .
git commit -m "Initial Cars CRUD application"
git remote add origin https://github.com/YOUR_USERNAME/cars-crud.git
git push -u origin main
```

Replace `YOUR_USERNAME` with your GitHub username. The included `.gitignore` keeps your virtual environment and SQLite database out of Git.

## 3. Create a branch for testing

```
git switch -c writing-tests
```

## 4. Write and run the tests

Follow TESTING_WORKSHOP.pdf. Complete the exercises for **unit, integration, API, and end-to-end (E2E)** testing: copy a working test, run it, make it fail on purpose, fix it, and write your own test.

After installing dependencies as explained in the workshop, verify the tests:

```
python -m pytest tests/unit tests/integration tests/api -q
python -m playwright install chromium
python -m pytest tests/e2e -q
```

**Requirement:** All tests must pass before submitting the pull request. Commit the restored, passing tests—not the intentional failures.

## 5. Push the testing branch

```
git add .
git commit -m "Add unit, integration, API, and E2E tests"
git push -u origin writing-tests
```

## 6. Create and merge a pull request

1. Open your GitHub repository and click **Compare & pull request** (or **Pull requests → New pull request**).
2. Select **base:** `main` and **compare:** `writing-tests`.
3. Title it `Add automated tests`, review the changed files, and create the pull request.
4. Once the tests are passing and the changes are reviewed, click **Merge pull request → Confirm merge**.

**Done when:** `main` contains the app and the new passing tests, and the pull request is marked **Merged**.